

3.44 Each rear tire on an experimental airplane is supposed to be filled to a pressure of 40 pounds per square inch (psi). Let X denote the actual air pressure for the right tire and Y denote the actual air pressure for the left tire. Suppose that X and Y are random variables with the joint density function

$$f(x, y) = \begin{cases} k(x^2 + y^2), & 30 \leq x < 50, \ 30 \leq y < 50, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find k .
- (b) Find $P(30 \leq X \leq 40 \text{ and } 40 \leq Y < 50)$.
- (c) Find the probability that both tires are underfilled.